

STAGE 1: GENERATIVE SYNTHESIS

Pipeline Operations:

- Latent space facial sampling
- Fractional-strided transposed conv
- Upsampling periodic interpolation

FLAW 1: SPECTRAL CHECKERBOARD

- Nyquist Dirac spikes at $(u_0, v_0) = (\pm 1/T_0, \pm 1/T_0)$
- Exponential attenuation under blur ($P \propto e^{-2\pi^2 \sigma^2}$)

STAGE 2: BOUNDARY COMPOSITION

Pipeline Operations:

- Convex hull landmark Delaunay warp
- Poisson gradient boundary integration
- Color match & boundary blending

FLAW 2: PHASE & GRADIENT MISMATCH

- Dirichlet boundary seam: $\lim_{\mathbf{x} \rightarrow \partial\Omega^-} \Delta f \neq \lim_{\mathbf{x} \rightarrow \partial\Omega^+} \Delta f$
- Spatial phase discontinuities along $\partial\Omega$

STAGE 3: CAMERA SENSOR RESIDUE

Pipeline Operations:

- Silicon sensor wafer capture
- PRNU noise modulation ($I = I_0(1 + K)$)
- In-the-wild container transcoding

FLAW 3: SENSOR PATTERN DEFICIT

- Synthetic face lacks authentic sensor PRNU
- Spatial noise variance discontinuity:
 $\text{Var}(\mathbf{I}_{\text{residual}})|_{\Omega} \ll \text{Var}(\mathbf{I}_{\text{residual}})|_{\Omega^c}$

DEFENSIVE FORENSIC INTEGRATION: DUAL-STREAM SPATIAL-FREQUENCY SYSTEM

Spatial Stream (ConvNeXt-Small) captures boundary Poisson seams | Spectral Tower (SRM + Bayar + FFT) captures Dirac spikes
SNR-Adaptive Gating ($\mathbf{g}_{\text{eff}} = \mathbf{g} \odot \gamma$) suppresses noise hallucinations | Bi-GRU models feature velocity deltas ($\Delta \mathbf{e}_t$)